

A Quick Review of Hypothesis Testing

Hypothesis tests allow us to answer simple “yes-or-no” questions, such as:

- Is the true coefficient β_j in a linear regression equal to zero?
- Does the expected blood pressure among mice in the treatment group equal the expected blood pressure among mice in the control group?

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Hypothesis testing proceeds as follows:

1. Define the null and alternative hypotheses
2. Construct the test statistic
3. Compute the p -value
4. Decide whether to reject the null hypothesis

1. Define the Null and Alternative Hypotheses

- We divide the world into *null* and *alternative* hypotheses.
- The null hypothesis, H_0 , is the default state of belief about the world. For instance:
 1. The true coefficient β_j equals zero.
 2. There is no difference in the expected blood pressures.

1. Define the Null and Alternative Hypotheses

- We divide the world into *null* and *alternative* hypotheses.
- The null hypothesis, H_0 , is the default state of belief about the world. For instance:
 1. The true coefficient β_j equals zero.
 2. There is no difference in the expected blood pressures.
- The alternative hypothesis, H_a , represents something different and unexpected. For instance:
 1. The true coefficient β_j is non-zero.
 2. There is a difference in the expected blood pressures.

2. Construct the Test Statistic

- The test statistic summarizes the extent to which our data are consistent with H_0 .

3. Compute the p -Value

- The p -value is the probability of observing a test statistic at least as extreme as the observed statistic, *under the assumption that H_0 is true.*

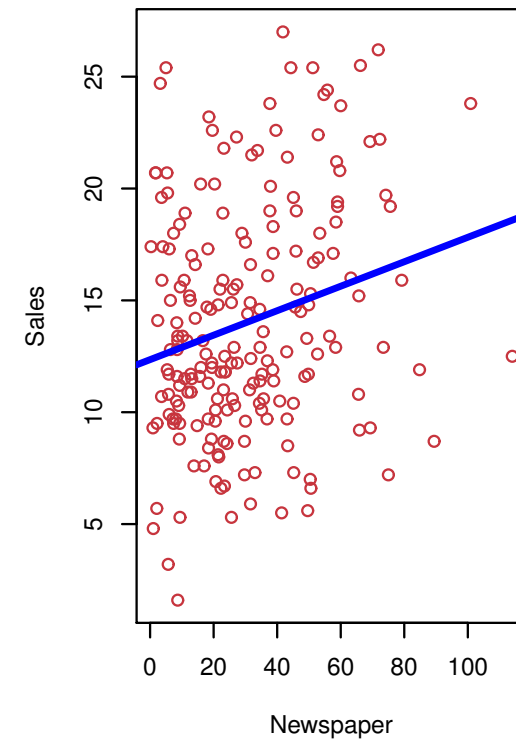
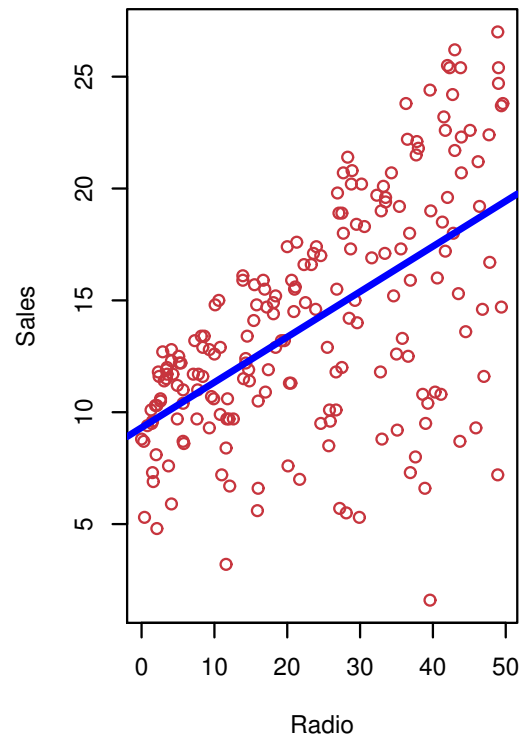
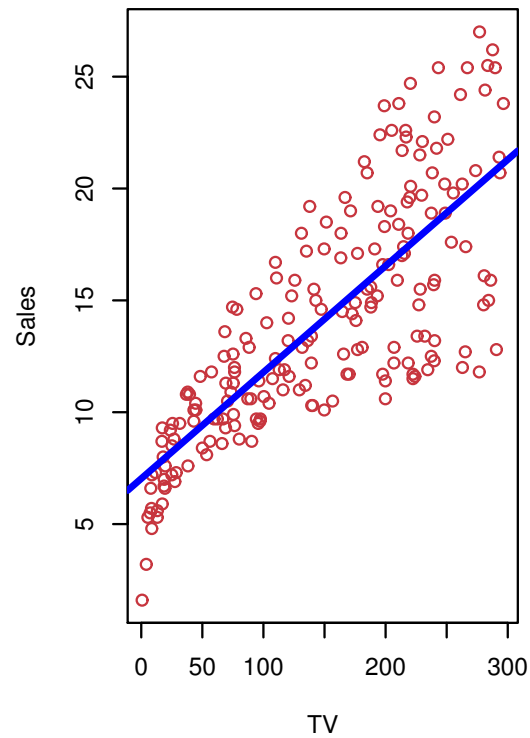
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- A small p -value provides evidence *against H_0* .

4. Decide Whether to Reject H_0 , Part 1

- A small p -value indicates that such a large value of the test statistic is unlikely to occur under H_0 .
- So, a small p -value provides evidence against H_0 .
- If the p -value is sufficiently small, then we will want to *reject* H_0 (and, therefore, make a potential “discovery”).

Advertising data



Results for the advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

Results for advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

Correlations:

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000